

DSA Revision Roadmap

30 Core Algorithmic Patterns • 180 Top Interview Problems

30
PATTERNS

180
PROBLEMS

Tracker Legend: Mark physical checkboxes upon completing each stage.

☐ Solve ☐ Code ☐ Rev 1 ☐ Rev 2 ☐ Rev 3

1. Sliding Window

6 Problems

ID	PROBLEM TITLE	DIFFICULTY	PROGRESS CHECKPOINTS
#3	Longest Substring Without Repeating Characters	Medium	☐ ☐ ☐ ☐ ☐
#76	Minimum Window Substring	Hard	☐ ☐ ☐ ☐ ☐
#209	Minimum Size Subarray Sum	Medium	☐ ☐ ☐ ☐ ☐
#424	Longest Repeating Character Replacement	Medium	☐ ☐ ☐ ☐ ☐
#567	Permutation in String	Medium	☐ ☐ ☐ ☐ ☐
#904	Fruit Into Baskets	Medium	☐ ☐ ☐ ☐ ☐

2. Two Pointers

6 Problems

#11	Container With Most Water	Medium	☐ ☐ ☐ ☐ ☐
#15	3Sum	Medium	☐ ☐ ☐ ☐ ☐
#16	3Sum Closest	Medium	☐ ☐ ☐ ☐ ☐
#18	4Sum	Medium	☐ ☐ ☐ ☐ ☐
#42	Trapping Rain Water	Hard	☐ ☐ ☐ ☐ ☐
#167	Two Sum II - Input Array Is Sorted	Medium	☐ ☐ ☐ ☐ ☐

3. Fast/Slow Pointers (Linked List)

6 Problems

#141	Linked List Cycle	Easy	☐ ☐ ☐ ☐ ☐
#142	Linked List Cycle II	Medium	☐ ☐ ☐ ☐ ☐
#19	Remove Nth Node From End of List	Medium	☐ ☐ ☐ ☐ ☐
#876	Middle of the Linked List	Easy	☐ ☐ ☐ ☐ ☐
#160	Intersection of Two Linked Lists	Easy	☐ ☐ ☐ ☐ ☐
#234	Palindrome Linked List	Easy	☐ ☐ ☐ ☐ ☐

4. Binary Search on Sorted Data

6 Problems

#33	Search in Rotated Sorted Array	Medium	☐ ☐ ☐ ☐ ☐
#34	Find First and Last Position of Element in Sorted Array	Medium	☐ ☐ ☐ ☐ ☐
#35	Search Insert Position	Easy	☐ ☐ ☐ ☐ ☐
#153	Find Minimum in Rotated Sorted Array	Medium	☐ ☐ ☐ ☐ ☐
#162	Find Peak Element	Medium	☐ ☐ ☐ ☐ ☐
#704	Binary Search	Easy	☐ ☐ ☐ ☐ ☐

5. Binary Search on Answer			6 Problems
ID	PROBLEM TITLE	DIFFICULTY	PROGRESS CHECKPOINTS
#875	Koko Eating Bananas	Medium	☐ ☐ ☐ ☐ ☐
#1011	Capacity To Ship Packages Within D Days	Medium	☐ ☐ ☐ ☐ ☐
#410	Split Array Largest Sum	Hard	☐ ☐ ☐ ☐ ☐
#774	Minimize Max Distance to Gas Station	Hard	☐ ☐ ☐ ☐ ☐
#1283	Find the Smallest Divisor Given a Threshold	Medium	☐ ☐ ☐ ☐ ☐
#1482	Minimum Number of Days to Make m Bouquets	Medium	☐ ☐ ☐ ☐ ☐
6. Hashing / Frequency Maps			6 Problems
#1	Two Sum	Easy	☐ ☐ ☐ ☐ ☐
#49	Group Anagrams	Medium	☐ ☐ ☐ ☐ ☐
#128	Longest Consecutive Sequence	Medium	☐ ☐ ☐ ☐ ☐
#217	Contains Duplicate	Easy	☐ ☐ ☐ ☐ ☐
#242	Valid Anagram	Easy	☐ ☐ ☐ ☐ ☐
#347	Top K Frequent Elements	Medium	☐ ☐ ☐ ☐ ☐
7. Prefix Sum / Running Sum			6 Problems
#303	Range Sum Query - Immutable	Easy	☐ ☐ ☐ ☐ ☐
#560	Subarray Sum Equals K	Medium	☐ ☐ ☐ ☐ ☐
#724	Find Pivot Index	Easy	☐ ☐ ☐ ☐ ☐
#930	Binary Subarrays With Sum	Medium	☐ ☐ ☐ ☐ ☐
#974	Subarray Sums Divisible by K	Medium	☐ ☐ ☐ ☐ ☐
#523	Continuous Subarray Sum	Medium	☐ ☐ ☐ ☐ ☐
8. Difference Array / Range Updates			6 Problems
#370	Range Addition	Medium	☐ ☐ ☐ ☐ ☐
#1094	Car Pooling	Medium	☐ ☐ ☐ ☐ ☐
#1109	Corporate Flight Bookings	Medium	☐ ☐ ☐ ☐ ☐
#1893	Check if All Integers in a Range Are Covered	Easy	☐ ☐ ☐ ☐ ☐
#1943	Describe the Painting	Medium	☐ ☐ ☐ ☐ ☐
#2381	Shifting Letters II	Medium	☐ ☐ ☐ ☐ ☐
9. Monotonic Stack			6 Problems
#739	Daily Temperatures	Medium	☐ ☐ ☐ ☐ ☐
#496	Next Greater Element I	Easy	☐ ☐ ☐ ☐ ☐
#503	Next Greater Element II	Medium	☐ ☐ ☐ ☐ ☐
#84	Largest Rectangle in Histogram	Hard	☐ ☐ ☐ ☐ ☐
#85	Maximal Rectangle	Hard	☐ ☐ ☐ ☐ ☐
#901	Online Stock Span	Medium	☐ ☐ ☐ ☐ ☐

10. Monotonic Queue / Deque			6 Problems
ID	PROBLEM TITLE	DIFFICULTY	PROGRESS CHECKPOINTS
#239	Sliding Window Maximum	Hard	☐ ☐ ☐ ☐ ☐
#862	Shortest Subarray with Sum at Least K	Hard	☐ ☐ ☐ ☐ ☐
#1425	Constrained Subsequence Sum	Hard	☐ ☐ ☐ ☐ ☐
#1438	Longest Subarray With Absolute Diff $\leq$ Limit	Medium	☐ ☐ ☐ ☐ ☐
#1499	Max Value of Equation	Hard	☐ ☐ ☐ ☐ ☐
#1696	Jump Game VI	Medium	☐ ☐ ☐ ☐ ☐
11. Heap / Top K			6 Problems
#215	Kth Largest Element in an Array	Medium	☐ ☐ ☐ ☐ ☐
#347	Top K Frequent Elements	Medium	☐ ☐ ☐ ☐ ☐
#692	Top K Frequent Words	Medium	☐ ☐ ☐ ☐ ☐
#703	Kth Largest Element in a Stream	Easy	☐ ☐ ☐ ☐ ☐
#973	K Closest Points to Origin	Medium	☐ ☐ ☐ ☐ ☐
#1046	Last Stone Weight	Easy	☐ ☐ ☐ ☐ ☐
12. Intervals			6 Problems
#56	Merge Intervals	Medium	☐ ☐ ☐ ☐ ☐
#57	Insert Interval	Medium	☐ ☐ ☐ ☐ ☐
#252	Meeting Rooms	Easy	☐ ☐ ☐ ☐ ☐
#253	Meeting Rooms II	Medium	☐ ☐ ☐ ☐ ☐
#435	Non-overlapping Intervals	Medium	☐ ☐ ☐ ☐ ☐
#452	Minimum Number of Arrows to Burst Balloons	Medium	☐ ☐ ☐ ☐ ☐
13. Greedy Scheduling / Sorting			6 Problems
#45	Jump Game II	Medium	☐ ☐ ☐ ☐ ☐
#55	Jump Game	Medium	☐ ☐ ☐ ☐ ☐
#406	Queue Reconstruction by Height	Medium	☐ ☐ ☐ ☐ ☐
#621	Task Scheduler	Medium	☐ ☐ ☐ ☐ ☐
#763	Partition Labels	Medium	☐ ☐ ☐ ☐ ☐
#134	Gas Station	Medium	☐ ☐ ☐ ☐ ☐
14. Linked List Manipulation			6 Problems
#21	Merge Two Sorted Lists	Easy	☐ ☐ ☐ ☐ ☐
#23	Merge k Sorted Lists	Hard	☐ ☐ ☐ ☐ ☐
#24	Swap Nodes in Pairs	Medium	☐ ☐ ☐ ☐ ☐
#25	Reverse Nodes in k-Group	Hard	☐ ☐ ☐ ☐ ☐
#92	Reverse Linked List II	Medium	☐ ☐ ☐ ☐ ☐
#138	Copy List with Random Pointer	Medium	☐ ☐ ☐ ☐ ☐

15. Tree DFS			6 Problems
ID	PROBLEM TITLE	DIFFICULTY	PROGRESS CHECKPOINTS
#104	Maximum Depth of Binary Tree	Easy	☐ ☐ ☐ ☐ ☐
#112	Path Sum	Easy	☐ ☐ ☐ ☐ ☐
#113	Path Sum II	Medium	☐ ☐ ☐ ☐ ☐
#543	Diameter of Binary Tree	Easy	☐ ☐ ☐ ☐ ☐
#124	Binary Tree Maximum Path Sum	Hard	☐ ☐ ☐ ☐ ☐
#226	Invert Binary Tree	Easy	☐ ☐ ☐ ☐ ☐
16. Tree BFS / Level Order			6 Problems
#102	Binary Tree Level Order Traversal	Medium	☐ ☐ ☐ ☐ ☐
#103	Binary Tree Zigzag Level Order Traversal	Medium	☐ ☐ ☐ ☐ ☐
#199	Binary Tree Right Side View	Medium	☐ ☐ ☐ ☐ ☐
#515	Find Largest Value in Each Tree Row	Medium	☐ ☐ ☐ ☐ ☐
#637	Average of Levels in Binary Tree	Easy	☐ ☐ ☐ ☐ ☐
#116	Populating Next Right Pointers in Each Node	Medium	☐ ☐ ☐ ☐ ☐
17. BST Problems			6 Problems
#98	Validate Binary Search Tree	Medium	☐ ☐ ☐ ☐ ☐
#99	Recover Binary Search Tree	Hard	☐ ☐ ☐ ☐ ☐
#230	Kth Smallest Element in a BST	Medium	☐ ☐ ☐ ☐ ☐
#235	Lowest Common Ancestor of a BST	Medium	☐ ☐ ☐ ☐ ☐
#450	Delete Node in a BST	Medium	☐ ☐ ☐ ☐ ☐
#700	Search in a Binary Search Tree	Easy	☐ ☐ ☐ ☐ ☐
18. Backtracking Basics			6 Problems
#46	Permutations	Medium	☐ ☐ ☐ ☐ ☐
#47	Permutations II	Medium	☐ ☐ ☐ ☐ ☐
#77	Combinations	Medium	☐ ☐ ☐ ☐ ☐
#78	Subsets	Medium	☐ ☐ ☐ ☐ ☐
#90	Subsets II	Medium	☐ ☐ ☐ ☐ ☐
#39	Combination Sum	Medium	☐ ☐ ☐ ☐ ☐
19. Backtracking with Constraints			6 Problems
#40	Combination Sum II	Medium	☐ ☐ ☐ ☐ ☐
#17	Letter Combinations of a Phone Number	Medium	☐ ☐ ☐ ☐ ☐
#79	Word Search	Medium	☐ ☐ ☐ ☐ ☐
#131	Palindrome Partitioning	Medium	☐ ☐ ☐ ☐ ☐
#51	N-Queens	Hard	☐ ☐ ☐ ☐ ☐
#52	N-Queens II	Hard	☐ ☐ ☐ ☐ ☐

20. Graph BFS / DFS			6 Problems
ID	PROBLEM TITLE	DIFFICULTY	PROGRESS CHECKPOINTS
#200	Number of Islands	Medium	☐ ☐ ☐ ☐ ☐
#695	Max Area of Island	Medium	☐ ☐ ☐ ☐ ☐
#733	Flood Fill	Easy	☐ ☐ ☐ ☐ ☐
#994	Rotting Oranges	Medium	☐ ☐ ☐ ☐ ☐
#1091	Shortest Path in Binary Matrix	Medium	☐ ☐ ☐ ☐ ☐
#1254	Number of Closed Islands	Medium	☐ ☐ ☐ ☐ ☐
21. Topological Sort / DAG			6 Problems
#207	Course Schedule	Medium	☐ ☐ ☐ ☐ ☐
#210	Course Schedule II	Medium	☐ ☐ ☐ ☐ ☐
#802	Find Eventual Safe States	Medium	☐ ☐ ☐ ☐ ☐
#1462	Course Schedule IV	Medium	☐ ☐ ☐ ☐ ☐
#1203	Sort Items by Groups Respecting Dependencies	Hard	☐ ☐ ☐ ☐ ☐
#2115	Find All Possible Recipes from Given Supplies	Medium	☐ ☐ ☐ ☐ ☐
22. Union Find / DSU			6 Problems
#547	Number of Provinces	Medium	☐ ☐ ☐ ☐ ☐
#684	Redundant Connection	Medium	☐ ☐ ☐ ☐ ☐
#1319	Number of Operations to Make Network Connected	Medium	☐ ☐ ☐ ☐ ☐
#1579	Remove Max Number of Edges to Keep Graph Traversable	Hard	☐ ☐ ☐ ☐ ☐
#990	Satisfiability of Equality Equations	Medium	☐ ☐ ☐ ☐ ☐
#1202	Smallest String With Swaps	Medium	☐ ☐ ☐ ☐ ☐
23. Shortest Path Algorithms			6 Problems
#743	Network Delay Time	Medium	☐ ☐ ☐ ☐ ☐
#787	Cheapest Flights Within K Stops	Medium	☐ ☐ ☐ ☐ ☐
#1514	Path with Maximum Probability	Medium	☐ ☐ ☐ ☐ ☐
#1631	Path With Minimum Effort	Medium	☐ ☐ ☐ ☐ ☐
#1334	Find City With Smallest Number of Neighbors	Medium	☐ ☐ ☐ ☐ ☐
#1976	Number of Ways to Arrive at Destination	Medium	☐ ☐ ☐ ☐ ☐
24. MST / Graph Greedy			6 Problems
#1584	Min Cost to Connect All Points	Medium	☐ ☐ ☐ ☐ ☐
#1135	Connecting Cities With Minimum Cost	Medium	☐ ☐ ☐ ☐ ☐
#1168	Optimize Water Distribution in a Village	Hard	☐ ☐ ☐ ☐ ☐
#1489	Find Critical Edges in Minimum Spanning Tree	Hard	☐ ☐ ☐ ☐ ☐
#778	Swim in Rising Water	Hard	☐ ☐ ☐ ☐ ☐
#1102	Path With Maximum Minimum Value	Medium	☐ ☐ ☐ ☐ ☐

25. Trie Data Structure			6 Problems
ID	PROBLEM TITLE	DIFFICULTY	PROGRESS CHECKPOINTS
#208	Implement Trie (Prefix Tree)	Medium	☐ ☐ ☐ ☐ ☐
#211	Design Add and Search Words Data Structure	Medium	☐ ☐ ☐ ☐ ☐
#212	Word Search II	Hard	☐ ☐ ☐ ☐ ☐
#648	Replace Words	Medium	☐ ☐ ☐ ☐ ☐
#677	Map Sum Pairs	Medium	☐ ☐ ☐ ☐ ☐
#1268	Search Suggestions System	Medium	☐ ☐ ☐ ☐ ☐
26. Bit Manipulation			6 Problems
#136	Single Number	Easy	☐ ☐ ☐ ☐ ☐
#137	Single Number II	Medium	☐ ☐ ☐ ☐ ☐
#191	Number of 1 Bits	Easy	☐ ☐ ☐ ☐ ☐
#338	Counting Bits	Easy	☐ ☐ ☐ ☐ ☐
#268	Missing Number	Easy	☐ ☐ ☐ ☐ ☐
#190	Reverse Bits	Easy	☐ ☐ ☐ ☐ ☐
27. 1D Dynamic Programming			6 Problems
#70	Climbing Stairs	Easy	☐ ☐ ☐ ☐ ☐
#198	House Robber	Medium	☐ ☐ ☐ ☐ ☐
#213	House Robber II	Medium	☐ ☐ ☐ ☐ ☐
#322	Coin Change	Medium	☐ ☐ ☐ ☐ ☐
#279	Perfect Squares	Medium	☐ ☐ ☐ ☐ ☐
#300	Longest Increasing Subsequence	Medium	☐ ☐ ☐ ☐ ☐
28. Knapsack / Subset DP			6 Problems
#416	Partition Equal Subset Sum	Medium	☐ ☐ ☐ ☐ ☐
#494	Target Sum	Medium	☐ ☐ ☐ ☐ ☐
#518	Coin Change II	Medium	☐ ☐ ☐ ☐ ☐
#474	Ones and Zeroes	Medium	☐ ☐ ☐ ☐ ☐
#1049	Last Stone Weight II	Medium	☐ ☐ ☐ ☐ ☐
#879	Profitable Schemes	Hard	☐ ☐ ☐ ☐ ☐
29. Grid Dynamic Programming			6 Problems
#62	Unique Paths	Medium	☐ ☐ ☐ ☐ ☐
#63	Unique Paths II	Medium	☐ ☐ ☐ ☐ ☐
#64	Minimum Path Sum	Medium	☐ ☐ ☐ ☐ ☐
30. String & Sequence DP			6 Problems
#1143	Longest Common Subsequence	Medium	☐ ☐ ☐ ☐ ☐
#72	Edit Distance	Medium	☐ ☐ ☐ ☐ ☐
#115	Distinct Subsequences	Hard	☐ ☐ ☐ ☐ ☐